CLASS 8 · CBSE · GANITA PRAKASH PART 1 · DPP SHEET

Chapter 1: A Square and A Cube

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Which of these cannot be perfect squares: 1057, 2304, 7928, 4000?
2. What is the units digit of 87^2 and of 126^2?
3. Given $60^2 = 3600$, find 61^2 by adding one odd number.
4. How many numbers lie between 31^2 and 32^2?
5. Find $\sqrt{2025}$ by prime factorisation.
6. Find the smallest number by which 2352 must be multiplied to get a perfect square, and the square root of the product.
7. Between which two consecutive whole numbers does $\sqrt{150}$ lie?
8. Find $\sqrt[3]{9261}$.
9. What is the smallest number by which 392 must be multiplied to get a perfect cube?
10. True or false: (i) a perfect cube can end in 8; (ii) every cube number has an odd number of factors.

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Answer Key

1. 1057, 7928 and 4000
2. 9 and 6
3. $3600 + 121 = 3721$
4. 62
5. $2025 = 3^4 \times 5^2$, so the root is $3 \times 3 \times 5 = 45$
6. $2352 = 2^4 \times 3 \times 7^2$; multiply by 3; root $= 84$
7. 12 and 13
8. $9261 = 3^3 \times 7^3$, so the cube root is 21
9. $392 = 2^3 \times 7^2$; multiply by 7
10. (i) True, for example $2^3 = 8$. (ii) False, for example 8 has 4 factors.

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